Multimodal AI for Talent Acquisition and Workforce Retention: An Integrated Decision-Support Framework

***Abstract*—High-throughput technical personnel recruitment entails complicated analysis with respect to candidate discovery, technical interview, and workforce retention. Traditional processes separate the mentioned analysis into distinct tasks in disconnected applications without an overall decision-making support solution. This paper introduces a multi-modal artificial intelligence system based on the five-levels modular architecture including twelve service containers. The technology stack includes a six-level BERT cross-encoder for candidate-job pairing, Faster-Whisper with voice activity detection for speech recognition, and a 12-layer DeBERTa-v3 fine-tuned for five-class short-answer scoring. The post-hire attrition risk is estimated via Optuna-tuned ordered CatBoost classifier with 52 tabular features, TreeSHAP explanations, and scenario analysis. Empirical evaluation on the modified benchmark dataset (1,470 instances; attrition prevalence rate 16.12%) via five-fold stratified cross-validation demonstrates that operating point equal to 0.33 improves minority-class recall by 15.16 percentage points (from 35.46% to 50.62%), F1 score - by 4.84 percentage points (from 45.50% to 50.34%), whereas ROC-AUC and PR-AUC metrics do not change (equal to 0.8071 and 0.5753, respectively). The three-level feature ablation test shows that introducing seven additional interaction variables causes the fold-level F1 score decrease of 3.85 percentage points ($p = 0.026$). The paper separates implemented and experimentally validated functionality and treats all prediction results as decision support advice under human responsibility.**

***Index Terms*—AI-assisted recruitment, semantic matching, short-answer grading, attrition prediction, feature ablation, explainable AI.**

I. Introduction

Hiring technical employees poses a number of complicated practical problems at different distinct evaluation stages. During recruitment, there are manual unstructured document screenings which rely on heuristic keyword approaches that fail to properly categorize semantically equivalent competencies. Technical candidates have multi-turn oral interviews but no consistent calibration scoring. Post-recruitment, there is generally no numerical method to predict future employee turnover risk that can incur high expenses related to recruiting, onboarding, production, and loss of knowledge.

Currently, there are no solutions that provide the ability to deal with all these hiring stages in one place: applicant-tracking systems provide lexical keyword screening, interviewing platforms lack the capability of automatic answer evaluation, and HR analytics dashboards are detached from hiring processes. Previous literature tends to treat the tasks of recruitment matching, interview assessment, and workforce attrition independently, thus motivating an architecture of integrated decision support systems based on empirical evaluation.

This work introduces a unified multimodal platform that integrates candidate intake, cross-encoder matching, speech recognition, calibrated short-answer grading, and tabular workforce attrition forecasting with the help of modular microservices architecture. This empirical study focuses mainly on workforce attrition prediction and feature ablation since this part of the platform has the most advanced and validated quantitative empirical evaluation.

The key contributions of this work are: 1) Integrated modular architecture: A five-tier modular framework that consists of twelve service containers that systemically integrate resume ingestion, cross-encoder matching, speech transcription, calibrated answer grading, and attrition forecasting. 2) Calibrated multimodal NLP pipelines: An ONNX-optimized 6-layer BERT Cross-Encoder (τ = 0.44) for candidate-job compatibility and a Platt-calibrated 12-layer DeBERTa-v3 pipeline for five-class short-answer assessment. 3) Reproducible attrition benchmark: A 5-fold stratified cross-validation study (N = 1,470) with comparison of seven tabular machine learning models with explicit operating-threshold analysis (τ = 0.33 vs. τ = 0.50) under harsh 1:5.2 class imbalance. 4) Empirical feature-parsimony result: Three-stage ablation showing the presence of statistically significant difference in F1 score (Δ = -3.85 percentage points, p = 0.026) at fold level when adding seven engineered interaction variables.

Three research questions motivate this study: RQ1: How do evaluated machine learning models vary in terms of threshold-dependent and threshold-independent metrics for attrition prediction? RQ2: Do progressively engineered features improve validated attrition prediction? RQ3: How to unify heterogeneous recruitment AI services into modular architecture while maintaining an explicit distinction between implementation and empirical evaluation?

II. Related Work

*Semantic Matching and Resume Screening.*The earliest form of resume screening was performed via lexical matching using TF-IDF and BM25. [1], Sensitivity to vocabulary mismatch exists for such measures of semantic match. [2] BERT provides context embeddings by considering dependencies within sequence pairs. Meanwhile, Sentence-BERT [3] allows similarity calculations between job description and profile through

efficient bi-encoder representation. Cross-encoders learn the complete interactions between tokens.

Automated Short-Answer Grading. Contextual language models have outperformed pure lexical automatic short-answer grading (ASAG). [4] Disentangled attention is used in DeBERTa, [5] while Platt scaling can be used to calibrate model outputs into rubric-based grades.

*Employee Attrition Prediction.*XGBoost [6], LightGBM [7], and CatBoost [8] known tree-based algorithms for tabular data containing categorical features. Predictive performance and explainability [9], [10]; recent attrition prediction algorithms have been combined in several turnover researches based on human resources data [11], [12]. Class imbalance, feature engineering and external validation issues are mentioned again in the systematic review on employee attrition. [13]. Imbalanced classification requires precision-recall analysis [14], and Optuna [15] supports hyperparameter optimization.

Explainability and Responsible AI. TreeSHAP [16] computes attribution for tree-based algorithms. Algorithmic fairness, disparate impact monitoring, and human supervision are important aspects for employment use cases. [17].

III. System Overview and Architecture

A. Four-Stage Recruitment Lifecycle

Fig. 1 illustrates the platform's end-to-end recruitment-intelligence lifecycle.

Stage 1 — Ingestion & Profiling. Resumes in PDF, DOCX, or TXT formats are parsed by the extraction service using (`pdf-parse`, `mammoth`) for structured text fields. A DeBERTa-v3-small regressor scores the resumes with respect to three criteria – organization, specificity, and completeness – on a scale from 1 to 100 based on 15,000 pseudo-labeled samples (Validation MAE: 5.65; Test MAE: 5.44). Those scores serve as heuristic guidance rather than gold standard annotations made by recruiters.

Stage 2 — Semantic Matching. Job descriptions and candidate descriptions are embedded into a joint space by a 6-layer ONNX-optimized BERT Cross-Encoder model (d=384, L≤256), to provide compatibility probability which is compared to the threshold τ=0.44. The matching module is functional, but its human-labelled independent ranking benchmark is incomplete.

Stage 3 — Interview Intelligence. Candidate's speech recorded with the use of Web Real-Time Communication (WebRTC) is split into segments by Silero VAD detector and transcribed with Faster-Whisper [18] (Tiny, INT8 CPU). The browser WebRTC deals exclusively with video output and audio capture without doing anything with faces and visual emotions. A 12-layer DeBERTa-v3 encoder evaluates the candidate's answers to interview questions based on rubrics with ordinal scale from 1 to 5; uncalibrated raw scores are Platt-calibrated. The protocol of human validation is established, but multi-rater validation results are missing.

Stage 4 — Retention Analytics. The feature vector of size 52 of a candidate profile triggers a pre-trained Optuna-tuned CatBoost classifier at τ=0.33, to return the probability of leaving the company and the risk level and the local TreeSHAP importance driver factors. Recruiters can use a What-If simulator to assess the difference in the risk $\Delta\hat{p}$ under policy simulation changes to represent conditional sensitivity of the model rather than causal evidence.

B. Layered System Architecture

Fig. 2 illustrates the system's modular structure in five layers, Table I describes all AI/ML components of the system along with their implementation and validation status. Express.js API Gateway on port (`:8080`) mplements JWT/RBAC security and routing logic. The platform includes twelve backend service containers with application/domain services and ML-specific services running in dedicated Python FastAPI containers on ports (`:4002`, `:4005`, `:4008--4010`). Application entities are stored in MongoDB with two logical databases (`lanka_talent`, `lanka_talent_analysis`).

IV. Methodology

A. Cross-Encoder Candidate Matching

Let $\mathbf{t}_{\text{job}}$ and $\mathbf{t}_{\text{cand}}$ stand for the text representations of job requisition and candidate profile (of length no more than 440 characters), respectively. The 6-layer BERT Cross-Encoder ($d = 384$, 12 heads) transforms the concatenated sequence $\mathbf{x}_{\text{pair}} = [\text{CLS}] \,\|\, \mathbf{t}_{\text{job}} \,\|\, [\text{SEP}] \,\|\, \mathbf{t}_{\text{cand}} \,\|\, [\text{SEP}]$ up to the maximum length $L \leq 256$. While dual-encoder systems estimate separate embeddings by means of cosine distance measure, the cross-encoder self-attention allows each token to interact with any other one regardless of whether it belongs to job requirements or candidate experience description. The output scalar logit z results in the match probability calculation as follows $P_{\text{match}} = \sigma(z) = (1 + e^{-z})^{-1}$. The criterion to be matched is $P_{\text{match}} \geq \tau_{\text{match}} = 0.44$.

B. Calibrated Interview Answer Scoring

Given question q, rubric reference r, and candidate response a, the inputs are created in the form of: Premise: $q \,\|$ "Expected: " $\|\, r$ and Hypothesis: "Candidate: " $\|\, a$. The 12-layer DeBERTa-v3 model ($L \leq 384$) generates probability vector $\mathbf{p} = [p_0, \ldots, p_4]^T$ of five ordinal classes: *Wrong*, *Poor*, *Average*, *Good*, *Excellent*. Having the validated class-score centers

$$S_{\text{cal}} = \text{clip}(0.9666\, S_{\text{raw}} + 0.0110,\ 0,\ 1) \tag{1}$$

The calibrated score is rescaled to a 0–100 point range, and scores are classified into ordinal tiers based on thresholds $[0.24, 0.47, 0.58, 0.84]$.

C. Employee Attrition Formulation

Let $\mathbf{x} \in \mathbb{R}^{52}$ represent an employee profile consisting of factors pertaining to employment, salary, satisfaction, and recruiting. The model predicts the probability of attrition $\hat{p} = P(y = 1 \mid \mathbf{x})$ where $y \in \{0, 1\}$. An Ordered CatBoost

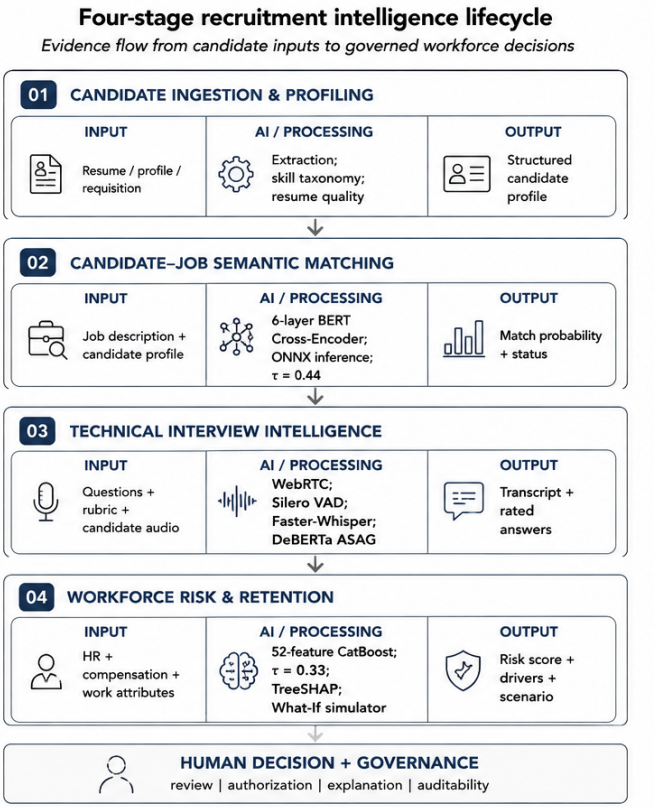


Fig. 1. Recruitment intelligence lifecycle.

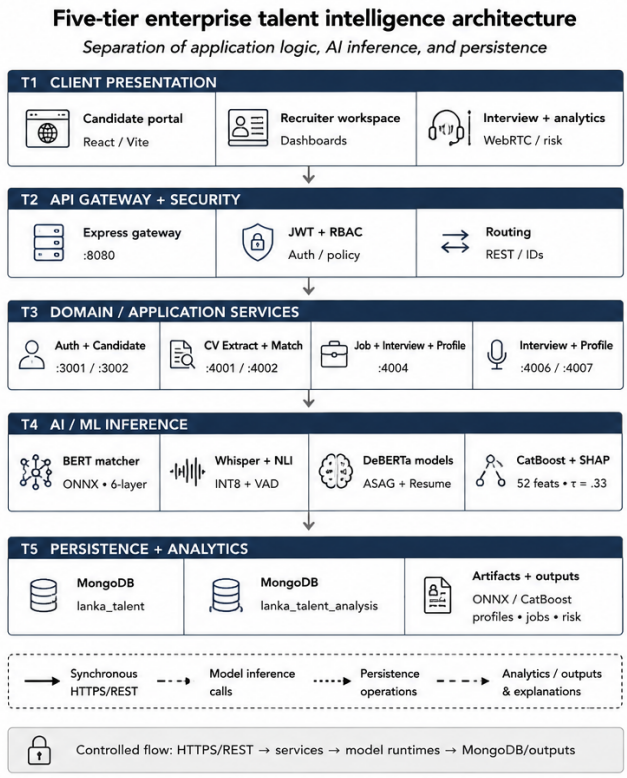


Fig. 2. Enterprise talent architecture.

TABLE I
INVENTORY OF AI/ML SUBSYSTEMS

Subsystem / Task	Core Architecture	Runtime	Input	Output	Status
Document Extraction	Regex parser (`pdf-parse`, `mammoth`)	Node.js :4001	PDF/DOCX/TXT	Structured sections	Operational
Skill Profile Scoring	DeBERTa-v3-small (3-target regressor)	FastAPI :4009	Context text	Quality scores (pseudo-labelled)	Impl. (pseudo-labels)
Candidate–Job Match	6-Layer BERT Cross-Encoder, ONNX	Node.js :4002	Seq. pair $L{\leq}256$	Prob. $\in [0,1]$; τ=0.44	Impl. (benchmark pending)
Audio Transcription	Faster-Whisper Tiny, INT8, Silero VAD	Python :4005	16 kHz PCM	Timestamped transcript	Operational
Answer Grading (ASAG)	12-Layer DeBERTa-v3 + Platt calibration	Python :4010	Triplet $L{\leq}384$	$S_{\text{cal}}{\in}[0,100]$, 5 class	Impl. (human val. pending)
NLI Concept Check	DeBERTa-v3-xsmall, zero-shot NLI	Node.js :4006	Premise–hyp.	Entailment score	Operational
Attrition Prediction	Ordered CatBoost, 52 feats, τ=0.33	FastAPI :4008	52-dim vector	Risk prob., binary class	Empirically validated
What-If Simulation	CatBoost re-scoring under perturbation	Node.js :4003	Modified feats	$\Delta\hat{p}$, risk tier	Operational

classifier is trained using minimization of weighted cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\left[w_{+}\, y_i \log \hat{p}_i + (1-y_i)\log(1-\hat{p}_i)\right] \qquad (2)$$

where the positive class weight equals w_{+}=1.3478. Optimal hyperparameter values (depth of tree $=$ 4, learning rate $=$ 0.0617, L_2 leaf regularization $=$ 12.02, 600 iterations) have been selected using a 50-iteration Optuna TPE search procedure. Binary classification uses an operating threshold $\tau_{\text{attr}} = 0.33$ so that $\hat{y} = \mathbb{I}(\hat{p} \geq 0.33)$. The feature attributions $\phi_j(\mathbf{x})$ computed using TreeSHAP [16], satisfying $\sum_{j=1}^{M}\phi_j(\mathbf{x}) = f(\mathbf{x}) - \mathbb{E}[f(\mathbf{x})]$.

D. What-If Scenario Simulation

Recruiters investigate the retention dynamics through what-if scenario simulation based on the trained models. For a given baseline $\mathbf{x}$, users change the candidate features (for instance, the salary offered, remote work type), resulting in a scenario vector $\tilde{\mathbf{x}}$. The service computes derived factors and evaluates the risk differential prediction $\Delta\hat{p} = \hat{p}(\tilde{\mathbf{x}}) - \hat{p}(\mathbf{x})$, thus estimating the model sensitivity to changed inputs instead of assuming causal effects in reality.

V. EXPERIMENTAL DESIGN

A. Provenance of Dataset and Feature Matching

The experiments evaluate the retention subsystem using a modified benchmark (N=1,470; Table II). The target variable indicates voluntary turnover with positive prevalence of (n=237 positive, n=1,233 negative), resulting in 1:5.20 class

TABLE II
ATTRITION DATASET

Property	Value
Total records (N)	1,470 (100%)
Attrition = 1 (positive class)	237 (16.12%)
Attrition = 0 (negative class)	1,233 (83.88%)
Class imbalance ratio	1 : 5.20
Raw dataset columns	53 (1 target, 52 attributes)
Excluded non-varying / IDs	4 (`EmployeeCount`, `Number`, `Over18`, `StdHours`)
Excluded protected demographics	3 (`Age`, `Gender`, `MaritalStatus`)
Base model input features	45 (26 standard IBM + 19 regional/hiring)
Standard IBM HR baseline	26 (Ablation Config A)
Regional & hiring attrs	19 (Added in Ablation Config B)
Engineered interaction variables	7 (Added in Ablation Config C)
Total production model features	52 (11 categorical, 41 numerical)
Missing values	0 (complete records)
Validation protocol	5-fold stratified CV (`random_state`=42)

imbalance and without any missing values. The raw tabular dataset consists of 53 columns including the binary target `Attrition` and 52 candidate attributes. Four constant or identifier attributes (`EmployeeCount`, `EmployeeNumber`, `Over18`, `StandardHours`) and three protected demographic attributes (`Age`, `Gender`, `MaritalStatus`) are excluded from the model inputs as a part of fairness-aware pre-processing procedure; however, such feature exclusion alone neither guarantees nor achieves fairness or avoidance of proxy discrimination. The set of 45 base features includes 26 standard IBM HR attributes (related to workplace, tenure, and satisfaction ratings) and 19 regional hiring attributes (including the compensation in Sri Lankan Rupees, working arrangements, location, and interview evaluations). With the introduction of seven new domain-specific engineered features, the total number of 52 model input features (11 categorical, 41 numerical) is reached.

B. Cross-Validation and Prevention of Data Leakage

All models are validated using 5-fold stratified cross-validation (`StratifiedKFold`, `random_state`= 42). In order to prevent the data leakage, all pre-processing steps including categorical encoding, numerical scaling and the threshold operation determination are limited only to the training folds. The operating threshold τ=0.33 was determined within-folds via maximization of balanced accuracy on the training-validation split during hyperparameter optimization.

C. Evaluated Models and Baselines

Five default algorithms are evaluated at the default threshold τ=0.50: Default CatBoost, XGBoost [6], LightGBM [7], Random Forest, and ElasticNet-egularized Logistic Regression. They are compared with Optuna-tuned Ordered CatBoost at the default threshold τ=0.50 and at the optimized threshold τ=0.33.

D. Evaluation Metrics for Imbalanced Learning

Due to the presence of 1:5.20 class imbalance, accuracy is an inappropriate metric because the simple majority classifier would reach 83.88% of accuracy with zero recall. Thus, the models are evaluated using threshold-based minority class metrics (Prec $=\frac{\text{TP}}{\text{TP+FP}}$, Rec $=\frac{\text{TP}}{\text{TP+FN}}$, BalAcc $=\frac{1}{2}$(Rec $+\frac{\text{TN}}{\text{TN+FP}}$) and F1 $=\frac{2\cdot\text{Prec}\cdot\text{Rec}}{\text{Prec+Rec}}$) in conjunction with threshold-independent discrimination curves (ROC-AUC and PR-AUC) [14].

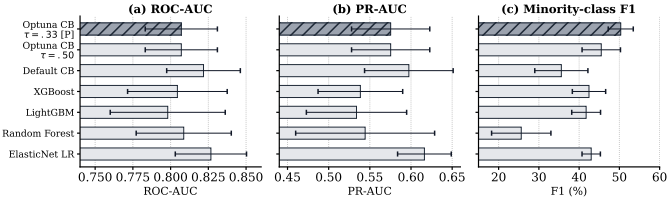


Fig. 3. Attrition model comparison.

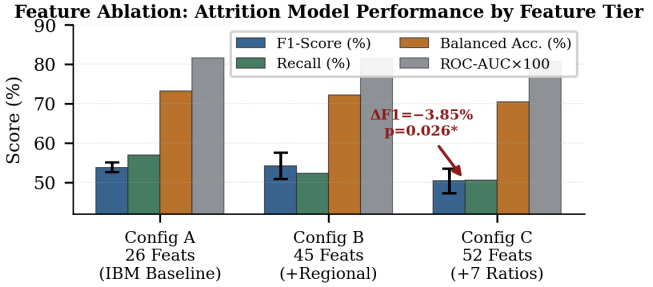


Fig. 4. Feature ablation results.

VI. RESULTS AND DISCUSSION

A. Benchmark Baseline and Operational Trade-Offs (RQ1)

Table III and Fig. 3 represent the benchmark 5-fold cross-validation results across all seven investigated model setups.

Regarding Threshold-Independent Discrimination. Examining continuous discrimination, ElasticNet Logistic Regression performs the best with the highest ROC-AUC (0.8267±0.0236) and PR-AUC (0.6163±0.0326). The second best in ROC-AUC terms is the Default CatBoost modelC (0.8218±0.0244), followed by Random Forest (0.8087) and Optuna CatBoost (0.8071). Regressed linear models perform well in continuous monotonic ranking but are not able to account for non-linear interactions and lack native handling of categorical variables. They are also not capable of localized What-If scenario simulations due to the exponentially growing dimensionality of the problem.

Minority-Class Suppression at Default Threshold (τ=0.50). Using the default decision threshold τ=0.50, all tree-based baselines demonstrate pronounced minority-class suppression. Random Fores model demonstrates a high level of precision (89.33±13.73%) but only identifies 15.19±4.91% of the departures (F1: 25.62±7.37%), while the Default CatBoost model identifies fewer than one out of four turnovers (recall: 23.22±5.67%, F1: 35.58±6.60%). In the context of technical hiring, false negatives (departures not identified and costing the company in recruitment, knowledge loss and onboarding costs) cause more disruption than false positives (advisory retention dialogue).

Efficacy of Operating-Threshold Optimization Adjusting the operating threshold to τ=0.33 in the Optuna CatBoost case

TABLE III
ATTRITION MODEL RESULTS

Model Configuration	τ	Acc (%)	Bal.Acc (%)	Prec (%)	Rec (%)	F1 (%)	ROC-AUC	PR-AUC
Optuna CatBoost [Proposed]	**0.33**	83.81±1.90	**70.40±1.50**	50.44±5.55	**50.62±2.72**	**50.34±3.11**	0.8071±0.0239	0.5753±0.0475
Optuna CatBoost	0.50	86.33±1.48	65.78±2.24	65.33±11.14	35.46±4.78	45.50±4.77	0.8071±0.0239	0.5753±0.0475
Default CatBoost	0.50	86.67±0.84	61.04±2.61	81.74±11.91	23.22±5.67	35.58±6.60	0.8218±0.0244	0.5974±0.0538
XGBoost	0.50	86.46±0.59	64.15±2.07	67.73±4.36	31.21±4.76	42.45±4.14	0.8045±0.0330	0.5385±0.0514
LightGBM	0.50	86.33±1.02	63.73±1.57	67.28±7.83	30.39±2.92	41.74±3.58	0.7981±0.0381	0.5338±0.0609
Random Forest	0.50	85.99±0.95	57.39±2.50	**89.33±13.73**	15.19±4.91	25.62±7.37	0.8087±0.0315	0.5443±0.0844
Logistic Reg. (ElasticNet)	0.50	**87.55±0.59**	63.95±0.94	82.54±6.75	29.12±1.71	43.00±2.28	**0.8267±0.0236**	**0.6163±0.0326**

TABLE IV
FEATURE ABLATION RESULTS

Config	#F	Acc (%)	BalAcc	F1 (%)	ROC	PR
A: IBM Baseline	26	84.22±0.90	73.21±1.20	53.78±1.26	0.816±0.020	0.583±0.040
B: +Regional	45	85.58±2.41	72.14±1.39	54.18±3.29	0.814±0.009	0.582±0.052
C: +7 Ratios	52	83.81±1.90	70.40±1.50	50.34±3.11	0.807±0.024	0.575±0.048
Δ: B−A	+19	+1.36	−1.07	+0.40 (p=0.83)	−0.002	−0.002
Δ: C−B	+7	−1.77	−1.74	−3.85* (p=0.026)	−0.007	−0.006

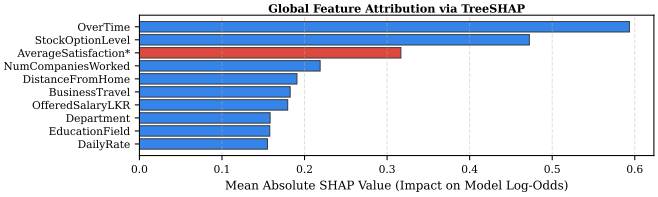


Fig. 5. Global SHAP feature importance.

TABLE V
SHAP FEATURE IMPORTANCE

Rank	Feature	$\overline{\lvert\text{SHAP}\rvert}$	Engineered?
1	`OverTime`	0.5935	No
2	`StockOptionLevel`	0.4723	No
3	`AverageSatisfaction`	0.3167	**Yes**
4	`NumCompaniesWorked`	0.2189	No
5	`DistanceFromHome`	0.1908	No
6	`BusinessTravel`	0.1826	No
7	`OfferedSalaryLKR`	0.1796	No
8	`Department`	0.1583	No
9	`EducationField`	0.1579	No
10	`DailyRate`	0.1551	No

allows balancing the precision–recall trade-off: recall of the minority class increases from 35.46±4.78% to 50.62±2.72% (+15.16 percentage points), F1 improves from 45.50±4.77% to 50.34±3.11% (+4.84 percentage points), and balanced accuracy is equal to 70.40±1.50%. Importantly, threshold-independent metrics (ROC-AUC: 0.8071, PR-AUC: 0.5753) are not affected by the change in the threshold as they only recalibrate the decision boundary but not the underlying model rankings. Therefore, these results suggest that ROC-AUC might be insufficient as the sole criteria of model selection under class imbalance.

B. Ablation and Feature Parsimony in Regularised Shallow Trees (RQ2)

Table IV and Fig. 4 show the feature ablation results using three nested sets of features with Optuna CatBoost setup (τ=0.33). Config A (26 baseline features from IBM) provides an F1 score = 53.78±1.26% and ROC-AUC of = 0.8160±0.0204. Adding 19 regional features (Config B, 45 features) slightly increases the accuracy to 85.58±2.41% and F1 to 54.18±3.29% ($\Delta = +0.40$ percentage points, paired t-test p=0.830), At the same time, regional information (local compensation LKR and work arrangements) is needed.

However, in case of adding 7 manually engineered interaction features (Config C, 52 features) F1 decreases to 50.34±3.11% ($\Delta = -3.85$ percentage points, paired t-test p=0.026, statistically significant under the defined fold-level paired test). Balanced accuracy similarly drops from 72.14% to 70.40%, and ROC-AUC goes down by 0.0071. This performance degradation is expected due to the presence of either redundancy or collinearity among the engineered features. However, the reason behind this performance decrease is not isolated in this study. Therefore, these results provide empirical support to the concept of feature parsimony in regularised shallow-tree ensembles.

C. SHAP Explainability and Scenario Simulation

Fig. 5 and Table V depict the top 10 global TreeSHAP attributions. `OverTime` received the highest average absolute attribution ($\overline{|\text{SHAP}|}$=0.5935), followed by `StockOptionLevel` (0.4723) and `AverageSatisfaction` (0.3167, the only engineered feature in the top 10). Regional compensation (`OfferedSalaryLKR`, rank 7, 0.1796) suggests that local salary scale is an important prediction factor.

It should be stressed that SHAP attributions estimate the significance of features within the decision space of the trained model, whereas causation between features and workforce turnover cannot be inferred from these values. Analogously, What-If tool provides predictions based on simulated changes of features but not causal effects of those changes.

VII. LIMITATIONS, THREATS TO VALIDITY, AND RESPONSIBLE AI

A. Limitations and Threats to Validity. *Dataset Scope:* Attrition experiments explore a modified benchmark (N=1,470); wider use across firms demands further validation, including cross-organizational and temporal holdouts. *Upstream Pipelines:* While BERT Cross-Encoder matching (τ=0.44)

and DeBERTa-v3 ASAG pipelines are operational, full human trials with multiple raters have not yet been completed; human-labeled benchmarks are deferred for future work. *Pseudo-Labels:* The DeBERTa-v3-small resume quality pipeline uses 15,000 rubric pseudo-labels for training (test MAE: 5.44), this shows consistency with heuristics but not with human annotations. *Validity:* The preprocessing and the choice of the threshold (τ=0.33) were rigorously separated from cross-validation folds to prevent any form of data leakage, yet the ordinal rubrics imply certain subjective design choices.

B. Responsible AI Governance. AI recruitment solutions raise significant ethical and legal questions, [17], handled via four types of governance control: 1) Human Decision-Making Control: AI scores and predictions serve only as advisory information in the decision-making process, shortlisting and hiring always being subject to human approval. 2) Model xplainability: The predictions of attrition include local Tree-SHAP explanations of feature contribution in the recruiter's interface. 3) Input Attributes Governance: The input attributes protected by law (gender, age, marriage) are explicitly prohibited from the set of model input attributes, whereas the disparate impact assessment among subgroups is recognized as a governance necessity. 4) Reducing Automation Bias: Predictions are provided to the user as advisory risk factors rather than determinants of employees' success.

VIII. Conclusion and Future Work

The presented paper presents an integrated multimodal AI approach to technical recruitment and retention analysis, built into a five-level modular architecture. This framework combines cross-encoders matching, voice transcription, calibrated short-answer grading, and retention prediction using CatBoost in an explainable decision support interface.

The comprehensive 5-fold stratified cross-validation performed on a modified 1,470-sample human resources dataset resulted in two main findings. The first finding is that the optimal tuning of the operating threshold value of (τ=0.33) improves the minority class recall by 15.16 percentage points compared to the default values (50.62% vs. 35.46%, F1: 50.34% vs. 45.50%) At the same time, ROC-AUC does not change and equals (0.8071). The second finding is that feature ablation shows that including 7 additional interaction features makes a statistically significant difference in F1 of the minority class at the fold level (-3.85 percentage points, p=0.026) under the specified paired test. TreeSHAP and What-If tool provide interpretable sensitivity analysis based on the model. The next step in the future research is conducting multi-rater interviews and building other ranking candidates benchmarks.

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